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Leakage-Controlled Resting-State Connectome Classification of Schizophrenia: A Nested Cross-Validation Audit of the COBRE Cohort

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Abstract. Resting-state functional connectomes are widely used to discriminate patients with schizophrenia from healthy controls, yet reported cross-validated accuracy depends heavily on the point at which features are defined relative to the train/test split. We audit this dependence on 146 participants of the COBRE cohort parcellated with the Schaefer-100 atlas (4,950 edges). Exploratory covariate-adjusted mass-univariate inference is kept strictly separate from prediction, and every predictive pipeline performs edge screening, dimensionality reduction and hyperparameter search inside the outer training folds of a repeated nested cross-validation. Under this regime, elastic-net logistic regression reaches an AUC of 0.768 ± 0.076 and PCA-30 an AUC of 0.771 ± 0.079, with an edges-only elastic net essentially unchanged at 0.765 ± 0.079. Fold-wise residualisation of mean framewise displacement lowers that edges-only estimate to 0.694. Holding a twenty-edge $|t|$ screen and a logistic classifier fixed, and varying only whether the screen sees the test fold, isolates leakage for that construction: AUC rises from 0.709 ± 0.076 to 0.903 ± 0.039. That 0.194 gap is the cost of leaking a hard univariate screen, not of leaking the primary elastic net. The same construction on a 50/50 age-, sex- and FD-matched subsample of UCLA CNP yields 0.679 against 0.873, again a 0.194 gap; that analysis is within-cohort nested cross-validation, not a locked transfer, and does not revise 0.77. On synthetic data with no signal the leakage-free protocol returns chance, whereas a leaked twenty-edge screen reports 0.82 when the edges are correlated as in a connectome and 0.94 when they are independent; the observed leaked value of 0.903 lies between those bounds and about 0.08 above the correlated floor. Calibration and subject-level aggregation of repeated out-of-fold predictions are reported for the leakage-free model. The resulting figure of roughly 0.77 AUC is specific to this cohort, parcellation, connectivity measure and model family rather than a general ceiling, and the audit protocol is released as public code.

Keywords: Functional connectivity · Cross-validation · Data leakage · Schizophrenia · Validation protocol

1 Introduction

Machine learning applied to functional neuroimaging is increasingly proposed as a route to computer-aided diagnosis in psychiatry [1, 2]. Schizophrenia is associated with altered resting-state functional connectivity, particularly across default-mode, salience and frontoparietal systems [3, 4, 5], and a full pairwise connectome supplies thousands of candidate features from a single scan. The apparent accuracy of a connectome classifier, however, is determined as much by the validation protocol as by the biology it is meant to capture.

A frequent pipeline screens edges on the whole cohort, retains the significant connections, and only then trains a classifier that is evaluated by cross-validation. Selection performed at that point encodes label information from participants who will later serve as test cases, and nesting the classifier alone cannot remove the resulting optimism [1, 6]. The distinction between exploratory group-level inference and predictive feature construction is therefore not a stylistic one: the two answer different questions and require different guarantees.

We quantify that distinction on the public COBRE cohort [7]. Our contributions are fivefold. First, we report a leakage-free benchmark in which elastic net, principal component analysis, connectome-based predictive modelling [8], per-fold univariate screening and network-level aggregation are all fitted inside the outer training folds of a repeated nested cross-validation. Second, we separate a covariate-adjusted exploratory map from every predictive pipeline. Third, we isolate the leakage effect with a contrast in which the selector, the classifier and the hyperparameter grid are identical and only the scope of the edge screen changes; this distinguishes optimism caused by leakage from optimism caused by a different model. Fourth, we examine head motion, probability calibration and the aggregation of repeated out-of-fold predictions, which jointly determine how the honest estimate should be interpreted. Fifth, we repeat the matched contrast within the UCLA CNP schizophrenia sample under a different preprocessor, and on signal-free synthetic connectomes. UCLA CNP is balanced by 1:1 nearest-neighbour matching on age, sex and mean FD (50/50) and is analysed with within-cohort nested cross-validation, not a train-on-COBRE transfer. Code and split definitions are publicly available.

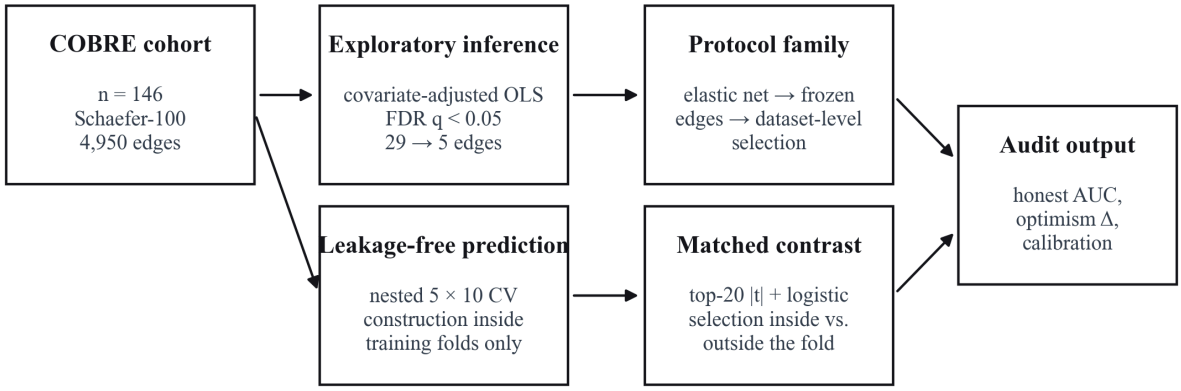


Fig. 1. Design of the audit on COBRE. Group-level inference and predictive modelling are kept separate. Every leakage-free pipeline fits its feature construction inside the outer training folds. Two comparisons then relax that constraint: a mixed protocol in which the feature space also changes, and a matched contrast in which only the scope of the edge screen changes.

2 Materials and Methods

2.1 Data and preprocessing

We used the publicly released COBRE resting-state fMRI collection (Figshare acquisition 4197885), preprocessed with the NIAK pipeline [7, 9]. Participants were retained when both the preprocessed BOLD volume and the accompanying confound time series were available, giving 146 individuals. The distributed phenotypic file provides age, sex and mean framewise displacement (FD); no symptom inventories were available, so the analysis concerns diagnostic classification only. All numeric confound regressors were passed to the Nilearn NiftiLabelsMasker during parcel-wise signal extraction with standardisation enabled [10]. Parcellation used the Schaefer 2018 atlas with 100 parcels under the seven-network Yeo solution at 2 mm [11, 12], resampled to each subject's BOLD grid by nearest-neighbour interpolation.

2.2 Connectivity estimation

Pearson correlations between all pairs of parcel time series produced one 100 × 100 symmetric matrix per participant. Discarding the diagonal and taking the upper triangle yields 4,950 undirected edges, which were Fisher *z*-transformed before modelling.

2.3 Exploratory group-level inference

Two edge-wise procedures were applied to the full cohort for descriptive purposes. Unadjusted two-sample *t*-tests with Benjamini–Hochberg control at $q < 0.05$ [13] identified 29 edges. A covariate-adjusted model, $FC_{ij} = \beta_0 + \beta_1 \cdot \text{diagnosis} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{sex} + \beta_4 \cdot \text{FD} + \varepsilon$, was then fitted per edge and the false discovery rate was controlled on β_1. These maps are reported as inference and were excluded from every leakage-free predictive pipeline.

2.4 Feature construction under fold isolation

Because feature construction rather than the classifier alone drives optimism, five strategies were compared, each fitted exclusively on outer training indices: an elastic-net logistic regression over all 4,950 edges; a 30-component PCA; connectome-based predictive modelling summarising the top and bottom decile of training-fold edge correlations into two scores [8]; the twenty edges with the largest training-fold absolute *t* statistic; and the 28 mean within- and between-network Fisher *z* values defined by the Yeo seven-network solution. Age, sex and FD were optionally appended as measured covariates [14]. Penalty paths and component loadings were never estimated on held-out participants.

2.5 Classifiers and nested validation

Classifiers were implemented in scikit-learn [15] and comprised L_2-penalised logistic regression, an elastic-net logistic regression with the saga solver, a radial-basis support vector machine and, for the network features, a random forest. Each was wrapped in a standardisation pipeline, and hyperparameters were chosen by an inner five-fold grid search maximising ROC-AUC. The outer loop was a stratified five-fold partition repeated ten times,

giving 50 outer splits. Fold-wise means and standard deviations are reported as descriptive summaries; because the repeats overlap, these standard deviations are not confidence intervals and are not used to rank methods against each other. Secondary analyses that required out-of-fold probabilities—subject-level aggregation, calibration, motion residualisation and the matched leakage contrast—used five folds repeated three times; the elastic-net estimate was stable between the two settings (0.770 versus 0.768).

2.6 Aggregation of repeated out-of-fold predictions

Under repeated cross-validation each participant contributes several test predictions. Concatenating all of them treats repeated observations from the same individual as independent. We therefore also computed a subject-level estimate in which the predicted probabilities of each participant are averaged first and a single ROC-AUC is evaluated over the 146 unique subjects. Both quantities are reported.

2.7 Leakage regimes

Two families of comparison were run. The first contrasts the leakage-free elastic net with a logistic regression restricted to the five covariate-adjusted edges of Table 2, whose identities come from full-cohort inference while only the classifier is cross-validated. Because the feature space and the classifier differ, that spread bounds mixed-protocol optimism and does not attribute it to leakage alone. A dataset-level selection figure inherited from the accepted submission is not reported here, because no released script regenerates it.

The second comparison isolates the selection step. The selector (the twenty largest absolute t statistics), the classifier (L_2 logistic regression), the covariate block and the hyperparameter grid are held fixed, and the only difference is whether the screen is computed within the outer training fold or on the complete cohort including the test fold. Label permutation used 50 shuffles of the leakage-free protocol with $p = (\#\{\mathrm{AUC}_{\mathrm{perm}} \geq \mathrm{AUC}_{\mathrm{obs}}\} + 1)/(n + 1)$; the smallest attainable value is $1/51 \approx 0.02$, which bounds the resolution of this test.

2.8 Motion sensitivity

Mean FD differs between groups, so three analyses probe its contribution. An edges-only elastic net omits the covariate block entirely. A residualisation variant regresses each edge on FD using coefficients estimated on the training fold and applies the same transformation to the test fold. Finally, five locked stratified 70/30 splits were evaluated, with all feature construction confined to the training portion.

2.9 Calibration and coefficient inspection

Expected calibration error over ten equal-width bins and the Brier score were computed from out-of-fold probabilities for the leakage-free models as well as for the leaked contrast. Coefficients of the leaked reference logistic model were aggregated across outer folds. Because that model draws on full-cohort edge identities and includes a group-imbalanced motion covariate, its weights are reported as a description of what the circular pipeline exploits and not as anatomical evidence.

2.10 Matched schizophrenia sample and null control

The matched contrast of Sect. 2.7 was repeated as nested cross-validation inside the UCLA Consortium for Neuropsychiatric Phenomics rest sample (OpenNeuro ds000030) [16]: 50 participants with schizophrenia who had an fMRIPrep MNI rest series, each paired with one control by greedy nearest-neighbour matching on standardised age, sex and mean FD (50/50). Time series used Schaefer-100 (4,950 edges); numeric confound columns entered the masker; the same three covariates were appended. This is within-CNP nested cross-validation under a different preprocessor, not a locked transfer from COBRE, and it does not revise 0.77.

The second extra experiment removes the signal. Labels are random and features are either independent Gaussians or connectomes built from random parcel time series with a five-component latent structure (mean absolute correlation between edge pairs 0.10 at the COBRE dimensions). For the CNP subsample we also permute the observed labels on the real matched connectomes, which preserves the empirical edge dependence. Ten draws were taken per configuration, including $k \in \{5, 20, 100\}$ and n between 50 and 600 at $p = 4{,}950$. A leakage-free protocol must return chance; whatever the leaked protocol returns is the level its own design can manufacture with no signal.

2.11 Implementation

Analyses were run in Python 3.11 with Nilearn, NiBabel, SciPy, statsmodels and scikit-learn [10, 15]. Connectivity matrices, edge statistics, benchmark tables and calibration summaries are exported as CSV. The audit protocol, the split definitions and the analysis scripts are available at https://github.com/Ishsirjan/Leakage_Audit; the imaging data remain at the original Figshare and OpenNeuro records.

3 Results

3.1 Sample and exploratory connectivity

Table 1 summarises the analysed sample. Patients moved more than controls during acquisition (Welch $t = 4.33$, $p = 3.3 \times 10^{-5}$). Unadjusted testing returned 29 FDR-significant edges; after adjustment for age, sex and FD only five survived (Table 2), a contraction consistent with the known influence of micro-movements on connectivity estimates [14]. The surviving connections couple visual, salience/ventral-attention, default-mode and control systems (Fig. 2), in line with large-scale network accounts of psychosis [4]. They are reported as exploratory inference and were withheld from the leakage-free pipelines.

Table 1. Characteristics of the 146 analysed COBRE participants. FD denotes mean framewise displacement in millimetres.

Group	*N*	*Age (years)*	*Male, n (%)*	*Mean FD*
Schizophrenia	72	38.2 ± 13.9	58 (80.6)	0.478 ± 0.274
Healthy control	74	35.8 ± 11.6	51 (68.9)	0.318 ± 0.152
Total	146	37.0 ± 12.8	109 (74.7)	0.397 ± 0.236

Table 2. Edges surviving false-discovery-rate control after adjustment for age, sex and mean framewise displacement. These edges are exploratory and were not supplied to any leakage-free predictive pipeline.

#	*Parcel 1*	*Parcel 2*	t_{adj}	q_{FDR}
1	LH Vis 3	RH Default Temp 2	4.38	0.031
2	LH SalVentAttn FrOperIns 1	RH Default PFCv 1	−4.93	0.011
3	LH Default Par 2	RH Cont PFCl 1	−4.30	0.031
4	RH Vis 8	RH Default Temp 2	−4.33	0.031
5	RH SalVentAttn TempOccPar 2	RH Default PFCv 1	−4.65	0.019

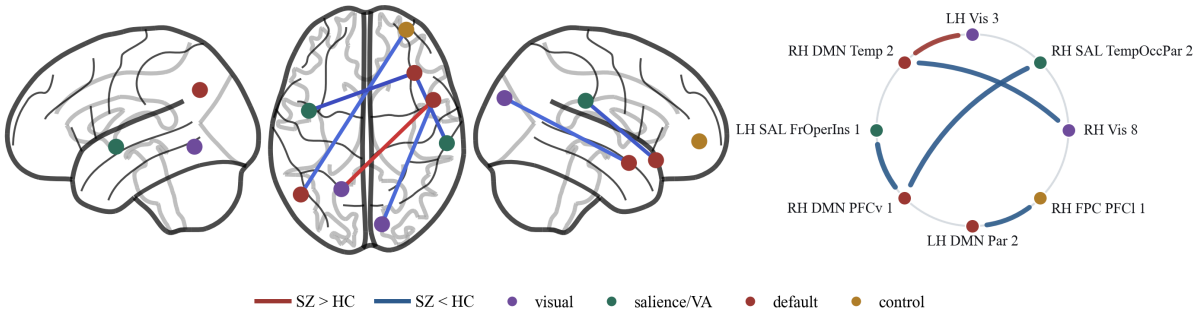


Fig. 2. Exploratory connectome of the five edges surviving covariate-adjusted FDR control, shown on Schaefer-100 centroids (left, sagittal and axial glass-brain projections) and as a chord diagram (right). The left sagittal view shows no edges because none of the five surviving connections is confined to the left hemisphere; three cross the midline (visible on the axial view) and two are right–right (visible on the right sagittal view). Warm edges indicate higher connectivity in patients, cool edges lower; line width scales with the adjusted t statistic. Node colour encodes the Yeo seven-network assignment.

3.2 Leakage-free classification

When no held-out information reaches feature construction, whole-connectome sparse and component-based representations perform best (Table 3, Fig. 3). PCA-30 with covariates reached 0.771 ± 0.079 and the elastic net 0.768 ± 0.076. Their fold distributions overlap almost completely, so we treat them as equivalent and designate the elastic net as the primary model because it returns sparse edge weights. Removing the covariate block changes

little (0.765 ± 0.079), which indicates that the imaging features rather than the appended demographics carry the discrimination. Hard univariate screening (0.705 ± 0.071) and network averaging (0.679 ± 0.092) performed clearly worse. Permutation of the labels placed the null mean near 0.49 with an empirical $p < 0.02$ at the resolution of 50 shuffles.

Table 3. Leakage-free benchmark over 50 outer splits (five folds, ten repeats). AUC and accuracy are fold means ± standard deviations; the last column pools all out-of-fold predictions. Standard deviations describe the spread across overlapping repeats and are not confidence intervals.

Feature construction	*Classifier*	*AUC*	*Accuracy*	*Pooled OOF AUC*
PCA-30 + covariates	Logistic	0.771 ± 0.079	0.713 ± 0.074	0.751
Elastic net, 4,950 edges + covariates	Elastic net	0.768 ± 0.076	0.687 ± 0.076	0.752
Elastic net, edges only	Elastic net	0.765 ± 0.079	0.682 ± 0.076	0.752
CPM, top/bottom decile + covariates	Logistic	0.742 ± 0.091	0.680 ± 0.083	0.733
Top-20 |t| within fold + covariates	Logistic	0.705 ± 0.071	0.641 ± 0.074	0.693
Network-28 + covariates	Logistic	0.679 ± 0.092	0.644 ± 0.079	0.670

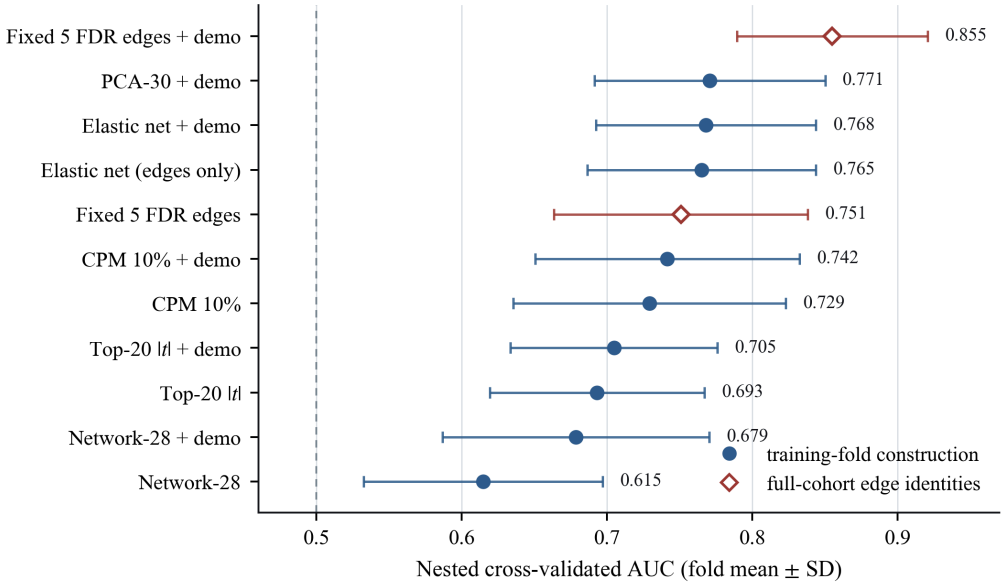


Fig. 3. Feature-construction strategies under nested cross-validation. Markers give the fold mean, whiskers one standard deviation across the 50 outer splits. Open diamonds mark the two settings whose edge identities were fixed on the full cohort and which are therefore not leakage-free.

3.3 Isolating the leakage effect

Relaxing fold isolation increases apparent performance (Fig. 4a). Freezing the five covariate-adjusted edges lifts the AUC to 0.855 ± 0.066 even though the classifier itself is cross-validated. That setting differs from the primary model in feature space and classifier as well as in selection scope, so the 0.087 spread is a mixed-protocol bound rather than leakage in isolation.

The matched contrast removes that ambiguity (Fig. 4b). With the same twenty-edge screen, the same logistic classifier and the same grid, computing the screen inside the training fold yields 0.709 ± 0.076, whereas computing it on the complete cohort yields 0.903 ± 0.039. The difference of 0.194 AUC is the cost of leaking this twenty-edge univariate screen, not the leakage cost of the primary elastic net, which already shrinks inside the fold and for which we do not report a matched leaked counterpart. The leakage-free arm here uses three repeats and matches the ten-repeat value of Table 3 (0.705) to within 0.004.

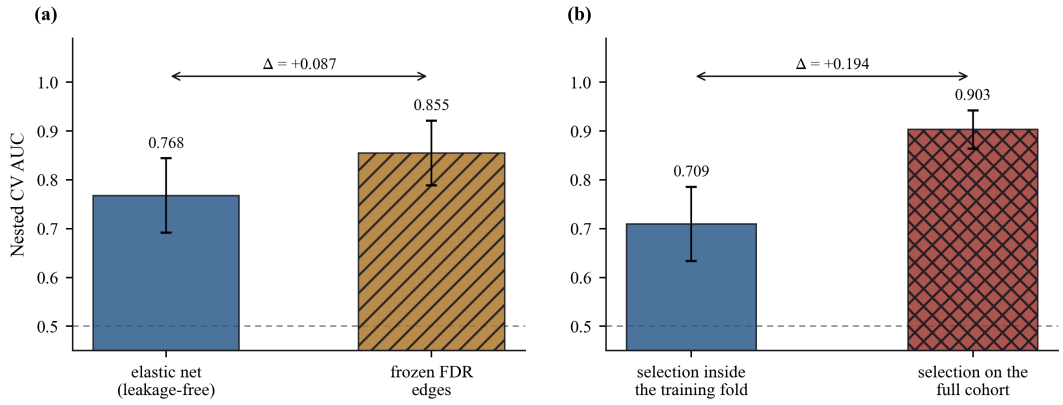


Fig. 4. (a) Mixed protocol: freezing the five full-cohort FDR edges raises apparent performance by 0.087 AUC, but the two settings also differ in feature space and classifier. (b) Matched contrast: identical top-20 screen, classifier and grid, differing only in whether the screen sees the test fold. The 0.194 AUC gap is the cost of leaking this univariate screen, not of leaking the primary elastic net.

3.4 Contribution of head motion

Motion is not eliminated by omitting FD from the design matrix. Residualising every edge on FD within the fold lowers the edges-only elastic net from 0.770 ± 0.072 to 0.694 ± 0.104 and the edges-only twenty-edge screen, which excludes the covariate block and therefore starts from 0.686 rather than the 0.709 of Sect. 3.3, to 0.647 (Fig. 5a). Locked 70/30 splits, in which feature construction never sees the evaluation set, gave a mean AUC of 0.733 for the elastic net and 0.712 for the univariate screen (Fig. 5b), consistent with the repeated nested estimate. Part of the honest signal therefore tracks diagnosis-related movement, and estimates obtained without motion control should be read accordingly.

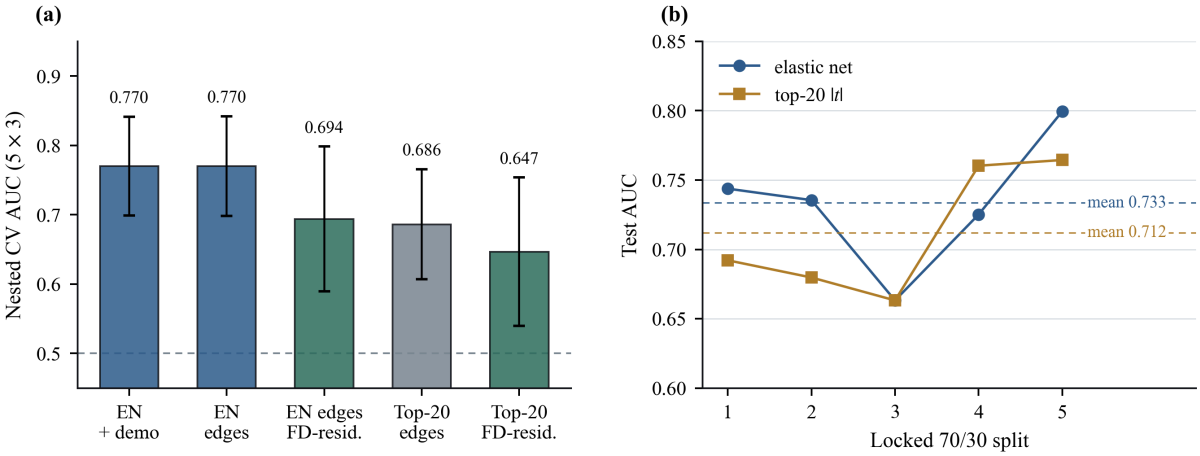


Fig. 5. (a) Removing mean framewise displacement from every edge within the fold lowers performance for both the elastic net and univariate screening, indicating that part of the honest signal is motion-related. (b) Five locked 70/30 splits with all construction confined to the training portion; dashed lines give the means.

3.5 Calibration and aggregation of repeated predictions

For the leakage-free elastic net the expected calibration error was 0.116 with a Brier score of 0.220 (Fig. 6a, Table 4); the model discriminates above chance but its probabilities would require recalibration before any decision-support use. The leaked arm of the matched contrast looks better on both counts, with an expected calibration error of 0.064 and a Brier score of 0.136. That improvement is itself an artefact: the leaked probabilities are fitted to edges chosen with knowledge of the very labels against which they are then scored, so calibration diagnostics inherit the leak and cannot be used to detect it. Good calibration is therefore not evidence that a pipeline is sound. Averaging the repeated out-of-fold probabilities within participants before computing a single AUC gave 0.764 for the primary model against 0.749 when all repeated predictions were pooled as if independent (Fig. 6b). The two aggregation rules differ by between 0.014 and 0.026 AUC across the regimes examined, so the choice does not alter the conclusions, but the subject-level value is the one we report.

Table 4. Aggregation of repeated out-of-fold predictions and calibration. Pooled AUC concatenates every fold prediction; subject-level AUC averages the repeated probabilities of each participant first. ECE is the expected calibration error over ten equal-width bins. The leaked row is better calibrated than the leakage-free rows, which shows that calibration cannot diagnose leakage.

Model	*Pooled AUC*	*Subject-level AUC*	*ECE*	*Brier*
Elastic net + covariates	0.749	0.764	0.116	0.220
Elastic net, edges only	0.749	0.764	0.118	0.220
Top-20 |t| inside the fold	0.705	0.731	0.080	0.221
Top-20 |t| on the full cohort	0.884	0.904	0.064	0.136

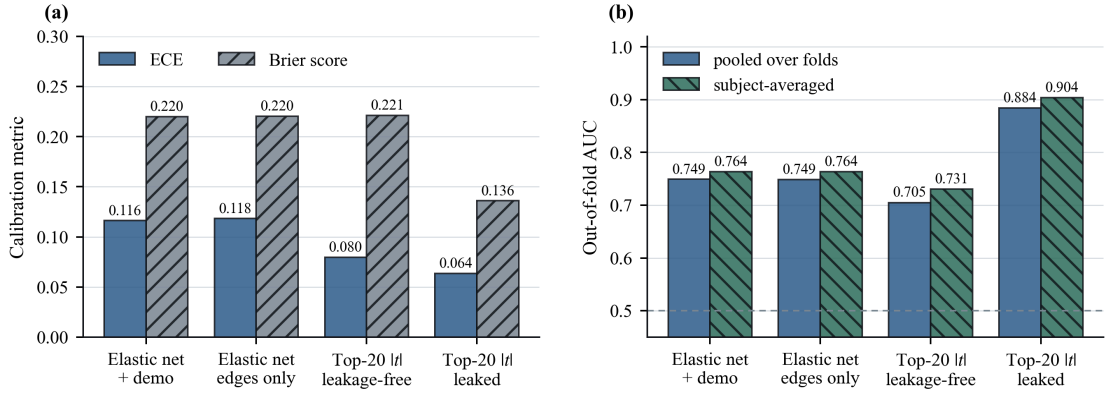


Fig. 6. (a) Calibration of the leakage-free models and of the leaked contrast. A leaked pipeline can be both more discriminative and better calibrated than an honest one, so neither metric detects the leak. (b) Pooling every repeated out-of-fold prediction as independent versus averaging the repeated probabilities within each participant before computing a single AUC.

3.6 What the leaked reference model uses

Aggregated across outer folds, the largest weight of the leaked reference model is mean FD (0.174 ± 0.125), followed by the left visual to right default-mode temporal edge (0.169 ± 0.103); all five covariate-adjusted edges recur among the leading contributors (Fig. 7). A motion summary outranking every connectivity feature in a model whose edges were selected on the full cohort is a description of the circularity, not a biological result, and we draw no anatomical inference from it.

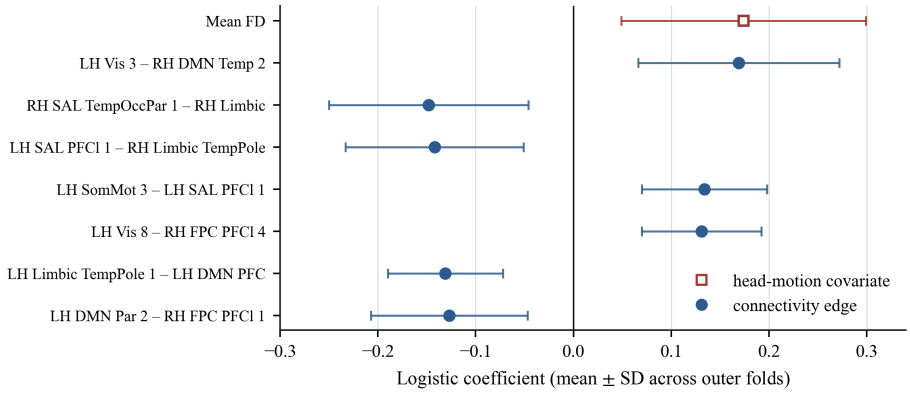


Fig. 7. Largest averaged coefficients of the leaked reference logistic model. Mean framewise displacement carries the largest weight, which characterises the circular pipeline rather than the neurobiology of schizophrenia.

3.7 Matched schizophrenia sample and null control

On a 50/50 age-, sex- and FD-matched UCLA CNP subsample the same twenty-edge construction yields 0.679 ± 0.074 inside the training fold and 0.873 ± 0.081 on the full subsample (Fig. 8b), a gap of 0.194 (five folds, three repeats). A leakage-free logistic model over all 4,950 edges reached 0.727. Accuracy is now interpretable (chance 0.50; leakage-free twenty-edge accuracy 0.617). fMRIPrep and the absence of a locked transfer mean these numbers do not replace 0.77; they show that the 0.194 inflation is not an artefact of COBRE's class balance.

The null control is the more informative extra experiment. On data with no signal the leakage-free protocol returns chance at the COBRE dimensions (0.505 ± 0.068 independent; 0.516 ± 0.064 correlated), which is what an

unbiased estimator must do and which validates the primary design. A leaked twenty-edge screen on the same data reports 0.821 ± 0.020 when edges are correlated as in a connectome and 0.938 ± 0.017 when they are independent. The leaked COBRE value of 0.903 therefore sits about 0.08 above the correlated floor and 0.03 below the independent ceiling (Fig. 8b): leakage accounts for most of that figure, but the correlated null does not swallow it entirely. Independent features are an upper bound on manufacturable optimism, not the reference for a real connectome. On the balanced CNP connectomes a leaked twenty-edge screen reports 0.873 on synthetic correlated nulls, 0.893 when the observed labels are permuted, and 0.971 on independent features. The observed leaked value of 0.873 therefore does not exceed the permutation floor: leakage accounts for that figure, while the leakage-free arm (0.679) remains above chance. Retaining more edges makes leakage worse: with independent features the leaked AUC rises from 0.809 at $k = 5$ to 0.937 at 20 and 0.998 at 100 (Fig. 8a). Optimism on null data falls with sample size, from 0.501 at $n = 50$ to 0.263 at 600 (Fig. 8c).

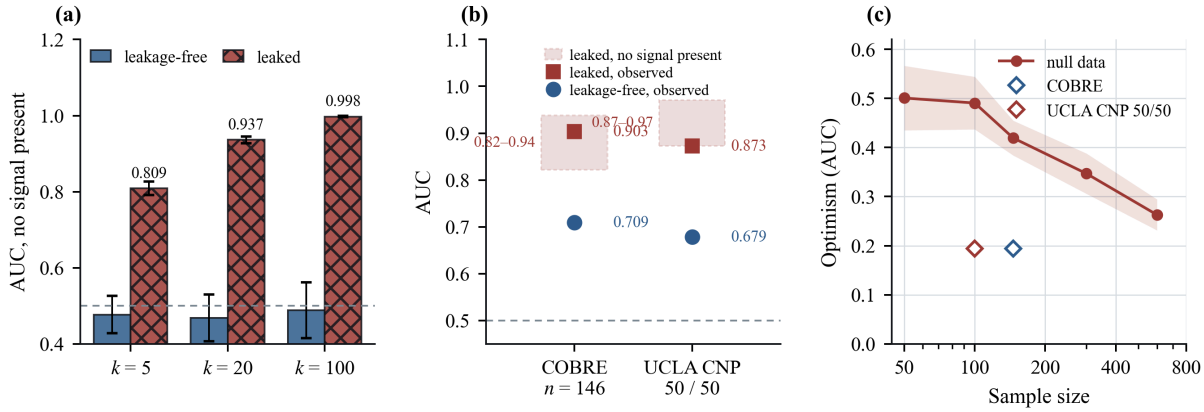


Fig. 8. (a) On data with no signal, a leaked twenty-edge screen reports more the more edges it retains, while the leakage-free protocol stays at chance. (b) Observed estimates against the band a leaked twenty-edge screen reaches on signal-free data of the same dimensions. The lower edge of each band uses correlated connectome edges and is the relevant floor; the upper edge uses independent features and is an upper bound. COBRE leaked (0.903) sits about 0.08 above the correlated floor. UCLA CNP is a 50/50 age-, sex- and FD-matched within-cohort audit under fMRIPrep, not a locked transfer; its leaked 0.873 sits at the correlated floor and does not exceed a label-permutation floor of 0.893. (c) Optimism against sample size on independent-feature null data.

4 Discussion

The central observation of this audit is that the same 146 scans support AUCs between 0.71 and 0.90 depending only on when a twenty-edge univariate screen is computed. That 0.194 gap is the price of leaking this construction. It is not the leakage cost of the primary elastic net, which already shrinks inside the fold. Freezing the five FDR edges raises the AUC to 0.855, but that comparison also changes the feature space, so it is not an isolated leakage effect.

The null control sharpens that reading. The relevant reference is a leaked twenty-edge screen run on signal-free connectomes with realistic edge dependence, which reports 0.821 at the COBRE dimensions. The leaked COBRE value of 0.903 exceeds that floor by about 0.08, so most of the leaked figure is a property of the design, but not all of it. Independent Gaussian features raise the same screen to 0.938 and should be read as an upper bound, not as evidence that 0.903 is indistinguishable from noise. A leaked figure should never be compared against chance; it should be compared against the same pipeline on label-shuffled or synthetic data of comparable dependence. The leakage-free arm returning chance on those data is what licenses reading 0.768 as signal.

The UCLA CNP contrast, after 1:1 matching on age, sex and FD, reproduces the same 0.194 gap (0.679 vs 0.873) and does not license reading 0.679 as a replication of 0.77.

The honest estimate itself should be scoped carefully. Roughly 0.77 AUC applies to this cohort, this parcellation, Pearson connectivity and this family of classical linear and sparse models; it is not a ceiling for schizophrenia decoding in general, and larger samples, different connectivity estimators or representation learning may well exceed it. Published COBRE results between 0.75 and 0.90 [1, 2] are best compared only after each study's selection protocol is known.

Feature construction mattered more than the choice of classifier. Regularised models over the full connectome and low-dimensional projections behaved similarly, while hard thresholding of univariate statistics lost about six AUC points. With about 117 training participants per split the ranking of individual edges is unstable, so methods

that accumulate many weak effects outperform those that discard them [6]. Since the fold distributions of the leading leakage-free methods overlap, we make no claim that any one of them is superior.

Motion deserves explicit treatment rather than a footnote. Mean FD differs between groups, dominates the weights of the circular model, and its fold-wise removal costs about 0.08 AUC. The edges-only model still discriminates, so the result is not reducible to movement, but a portion of the honest performance is motion-related and this should be stated whenever connectome classifiers are compared across cohorts with different acquisition quality.

Several limitations remain. UCLA CNP is a within-cohort fMRIPrep audit on a 50/50 matched subsample, not a locked transfer, and has no fold-wise FD residualisation of edges. The correlated synthetic null is a five-component latent structure; the tighter CNP control is a label permutation of the observed matched connectomes. Connectivity is Pearson correlation, the model family is classical, the permutation test on COBRE resolves probabilities only to about 0.02, and secondary analyses use three rather than ten repeats. No motion-matched COBRE subsample was constructed.

5 Conclusion

We measured how well resting-state connectomes classify schizophrenia in COBRE when feature construction is denied access to test labels, and how much that estimate inflates when access is granted. Under repeated nested cross-validation the honest AUC is 0.768 ± 0.076, while a twenty-edge screen that sees the full cohort reports 0.903, about 0.08 above the same screen on signal-free connectomes. Four practices follow for connectome classification: validate the feature-construction step with the same rigour as the classifier; report leakage as a matched contrast in which only the selection scope changes; compare any leaked figure with a synthetic or permuted null of comparable feature dependence, not with chance; and keep exploratory group maps separate from predictive features. The protocol and all scripts are available at https://github.com/Ishsirjan/Leakage_Audit.

Disclosure of Interests

The author has no competing interests to declare that are relevant to the content of this article.

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